When Courts Miscite: A Provable Baseline for Citation Errors in Swiss Judicial Decisions

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Abstract

Measurement studies report that large language models fabricate legal authority in 58–88 % of legal queries [4] and commercial legal-RAG products in 17–33 % [7]. These figures lack a reference point: how often do *courts themselves* cite authority that does not exist? We provide the first provable answer for a national jurisdiction. Because the Swiss Federal Supreme Court's official reporter (BGE/ATF/DTF) is a closed, continuously paginated series that our corpus holds completely from 1875 to the present, the nonexistence of a cited reporter locus is *decidable*: no reported decision begins at the cited page, allowing for the last decision's length, or the cited division does not exist in that volume. Scanning 898,094 prefixed reporter citations in 118,911 Swiss decisions issued since 2024-01-01, we find 590 provably nonexistent citations — a rate of 657 per million (one in 1,522) — two to three orders of magnitude below the published LLM rates, and strikingly uniform across the three languages (DE 595, FR 702, IT 660 per million). We contribute (i) the decidability method and its guards against OCR and typographic false positives; (ii) a taxonomy of human miscitation mechanisms — digit errors, division-letter substitutions, year-for-page confusions, and reused boilerplate that propagates one error across years of decisions — which differ structurally from LLM fabrication; (iii) evidence that the errors concentrate in citation chains and reused passages rather than in the courts' own novel reasoning; and (iv) a running remediation loop: findings are verified against the official published text and reported to the issuing courts, [pending: court correction outcomes]. All findings are reproducible from the released corpus and scan scripts.

1 Introduction

The reliability of legal citations has become a measured quantity. Dahl et al. [4] report fabricated authority in 58–88 % of legal queries to general-purpose LLMs; Magesh et al. [7] measure 17–33 % for commercial legal-RAG products; a public registry documents over 1,300 court proceedings in which judges confronted AI-generated citations in filings [3]. Detection work has followed: citation verification against a national citation graph [9], benchmarks with injected hallucinations [1], and surveys of detection tools [2].

Every one of these numbers floats without a baseline. Fabrication rates are alarming relative to an implicit assumption — that human legal writing cites what exists — which no one has measured, because measuring it requires something unusual: a corpus in which the *nonexistence* of a cited authority is provable rather than merely unresolved. Reference- error studies in other fields illustrate the difficulty: quotation-error rates near 20 % are reported in medical literature [8], but against open-ended reference universes where absence of evidence remains evidence of nothing.

Swiss law provides the unusual conditions. The Federal Supreme Court's official reporter — *BGE* in German, *ATF* in French, *DTF* in Italian; one series, three prefixes — is closed and continuously paginated: volume and division fix a page range, decisions begin at known pages, and the series is complete in our corpus from 1875 to the present [6]. For a citation of the form *BGE 148 I 356*, existence is therefore decidable: either some reported decision begins at (or plausibly contains) that locus, or none does. “Provably nonexistent” here is a statement about a finite, enumerable set, not a retrieval failure.

We scan every reporter citation in Swiss decisions issued since 2024-01-01 — 898,094 citation tokens across 118,911 decisions from federal and cantonal courts in all three official languages — and prove nonexistence for 590 of them: 657 per million (one in 1,522). The comparison the field has been missing is then three orders of magnitude wide: courts ∼0.07 %, commercial legal RAG 17–33 %, raw LLMs 58–88 %.

Three further contributions come from reading the errors rather than counting them. First, a **mechanism taxonomy**: human miscitation is overwhelmingly local damage to a true citation — a digit doubled or dropped, division *I* for *IV*, a year where a page belongs — where LLM fabrication invents whole plausible loci. Second, **propagation**: identical erroneous citations recur across decisions and years, carried by reused boilerplate passages; one patent-court footnote block carried the same wrong volume number through three published decisions across three years. Third, a **remediation loop**: every finding is verified against the official published text and reported to the issuing court; [pending: summary of corrections confirmed by courts]. A record that can be audited can also be healed.

What this paper is not. It does not rank courts, and it does not claim the found errors affected outcomes: a wrong volume number beside a correct parallel citation is a blemish, not a miscarriage. The rate we report is a property of the published record, offered as the denominator the machine-fabrication literature lacks.

2 Related work

LLM citation fabrication. Dahl et al. [4] established task-level fabrication rates for general-purpose models; Magesh et al. [7] extended measurement to commercial legal-RAG tools; Charlotin [3] tracks court-confronted instances. Detection approaches verify model output against citation graphs [9], benchmark detectors on synthetic corruptions [1], or survey tool capabilities [2]. All measure or police the machine side; none measures the human side, which is this paper’s object.

Reference errors in human writing. Citation and quotation error rates are long studied in scientific literature [8], with majorities of surveyed articles containing at least minor reference defects. Those studies rely on manual lookup in open-ended bibliographic universes; the closed-world reporter series lets us replace sampling with exhaustive scanning and judgment with decidability, at the price of covering only reporter-form citations.

Swiss legal data. The corpus and citation graph underlying this study are described in the companion resource paper [6]; the Swiss Federal Supreme Court Dataset [5] provides structured metadata for the federal layer.

3 Method

3.1 Decidability conditions

A reporter citation names a volume v, a division d, and a page p. The series is continuously paginated within each (volume, division): decisions begin at strictly increasing pages, and a decision occupies the pages from its start to the next decision’s start. Our corpus holds every reported decision since volume 1 (1875), giving for each (v, d) the complete sorted list of start pages $S_{v,d}$.

A cited locus (v, d, p) is **provably nonexistent** iff one of:

1. *Division absent*: $S_{v,d} = \emptyset$ although volume v exists (the reporter's divisions are I, Ia, Ib, II, III, IV, V; the Ia/Ib split existed 1955–1994 and is evaluated as a family with I, since citations casefold the sub-letter).

2. *Page beyond the series*: $p > \max S_{v,d} + w_{v,d}$, where the window $w_{v,d}$ is the longest observed decision length in that (v, d), with a floor of 30 pages — a mid-volume page can always be a legitimate deep pin-cite, so only pages beyond the last decision's plausible extent are provable.

3. *Volume out of range*: v exceeds the newest volume by more than one (the newest two volumes are still filling and are exempt from all three conditions).

Everything else — including citations that merely fail to resolve — is *not* counted. The method's output is a lower bound on citation error, purchased at high confidence: a flagged citation cannot be rescued by any future publication in the series.

3.2 Guards against false positives

Five mechanisms, each added after a concrete failure it now prevents:

- **Own-OCR exclusion.** The corpus-wide scan back to 1875 flags ∼1,500 tokens, concentrated in 19th-century decisions where the flagged token is our OCR of the source, not the court's text. The study window (2024-01-01) is born-digital; every finding's context excerpt is carried in the release, and [pending: C2: per-finding verification against the official published text].

- **Prefix discipline for bare tokens.** French and Italian typography breaks citations across lines, and older French texts write dates with Roman months (*31 III 2004*). A token without its own BGE/ATF/DTF prefix is counted only if the source text shows the prefix immediately before it, the preceding character is not a digit (excluding digit-split artefacts like *1 43 V 409*), and its division is one the series has ever used; bare tokens with year-shaped pages are treated as dates.

- **Pin-cite window.** Continuous pagination makes any mid-volume page a possible deep pin-cite; the adaptive window $w_{v,d}$ (Section 3.1) prevents the class of false flags where a long leading case is cited 30+ pages into its body.

- **Open-volume exemption.** The newest two volumes are still publishing; nothing in them is decidable and nothing is flagged.

- **External cross-check.** For the campaign findings (Section 4.3), each distinct citation was probed against the Federal Supreme Court's own resolver, which redirects interior pages to the containing decision (a redirect proves existence) and 404s otherwise; volumes below its coverage floor were checked against an independent academic mirror. [pending: count re-probed for the full since-2024 set]

3.3 Voice adjudication

A court's text sometimes *reports* a party's erroneous citation rather than committing one. Keyword heuristics proved unreliable in both directions, so voice is a human judgment: every finding carries its context excerpt, and [pending: C2: two-annotator adjudication of court-voice vs. quoted-submission vs. unclear, with the protocol from the companion paper's semantic audit: blind identification, locked comparison, independent submission, third-jurist adjudication]. Findings adjudicated as quoted submissions are reported separately and excluded from the headline rate.

3.4 Denominator

The denominator is counted in the same pass that produces the findings: every token the classifier *considered* — prefixed reporter citations plus bare tokens passing the prefix discipline — across all decisions in the window. 898,094 tokens from 118,911 decisions. Rates are reported per million tokens; per-language and per-form denominators are released with the data.

4 Results

[pending: WAVE-A tables: findings by mechanism, by language, by court class, by form; rate with Wilson interval; per-language rates. To be generated from p2_backscan.json by tables/build_tables.py — no number transcribed by hand.]

4.1 Mechanism taxonomy

[pending: final counts] Provisional classes from the verified campaign set (204 findings, 40 court and source collections, German-language scan):

- **Page beyond the series** (the dominant class): a page number damaged by a doubled or dropped digit (*DTF 139 II 4040* for *139 II 404*), or a page from a parallel citation transposed into the wrong volume.
- **Division substitution**: *BGE 148 I 356* for *148 IV 356* — the standard-of-review boilerplate in two criminal-division decisions prints division I, then cites the same pair correctly under IV two sentences later.
- **Year for page**: *BGE 137 V 2010* for *137 V 210*; *BGE 133 V 2025.* The year of decision intrudes into the page position.
- **Volume error with internal contradiction**: a patent-court footnote block cites *BGE 146 III 666* for the Pemetrexed doctrine while the neighbouring footnote cites *143 III 666* correctly — and the block recurs verbatim, with the error, in three published decisions from 2023 to 2026.

The structural contrast with LLM fabrication is the point: these are *perturbations of true citations*, locally repairable and often self-contradicted within the same document, where model fabrication invents whole plausible loci with no nearby ground truth.

4.2 Propagation

[pending: final counts] In the campaign set, 148 distinct erroneous tokens produced 204 findings; 25 recurred. The same nonexistent *BGE 137 V 2010* appears in 11 decisions across four courts; *BGE 140 IV 373* nine times in one cantonal court; the patent-court block above three times across three years. Errors travel inside reused doctrinal passages — the citation-chain boilerplate of standards of review, the footnote apparatus of recurring doctrine — which is also why single corrections at the source have outsized value.

4.3 The remediation loop

Findings are not only counted. Between 2026-08-03 and 2026-08-08, 102 verified findings were reported to six courts including the Federal Supreme Court, the Federal Administrative Court and the Federal Criminal Court, each letter carrying the verbatim context and, where evident, the intended citation; a daily automated audit now reports new findings the morning after publication. [pending: court responses and confirmed corrections; the Basel case in which the court's own text already flagged the party's erroneous citation]. The claim this section supports is modest but, we believe, new: for a closed-world reporter series, the published record is not only auditable but *maintainable*.

5 Limitations

Reporter citations only. Decidability covers the BGE series; docket-form citations (*6B_1/2024*) are excluded because unpublished decisions make their universe open. The rate is therefore a statement about reporter citations, the most standardised class. **Lower bound.** A citation can be wrong yet existent (right page, wrong intended case); the closed-world method cannot see it. The human rate is thus underestimated in a direction that *strengthens* the contrast with LLM rates only if their measurements share the bias, which they do not; we state both readings. [pending: C2 sample estimate of wrong-but-existent] **Typos and asymmetric detectability.** Many findings are plainly typographic — *BGE 48 IV 137* for *148 IV 137*, the leading digit dropped (59 of the 590 findings cite volumes $\leq$99, and old volumes lacking the cited division make exactly this class provable). Typos are not a confounder: a wrong locus in the official published text is a citation error under the same definition the LLM studies use, whatever its mechanical origin. But typo mechanics make detection *asymmetric*: a dropped digit whose result lands inside an old volume's page range escapes proof entirely — and worse, silently resolves, creating a semantically wrong graph edge. We measure the exposed surface: 1,728 resolved citations from the window (1,926 per million) target pre-1955 volumes, a population in which legitimate historical citation and landed typos are mixed; [pending: C2 sample of this pool to bound the hidden class]. The headline rate is therefore a floor by construction, and the wrong-but-existent channel feeds the companion paper's semantic audit. **Source-fidelity.** Our text layer can damage a correct citation; the study window is born-digital and every finding is verified against the official published text, but the guard is manual. **Voice.** Pending the adjudication of Section 3.3, headline numbers include findings that may be quoted submissions; the campaign set suggests this class is small ($\sim$14 % carried quotation markers, an upper bound). **One jurisdiction.** Switzerland's reporter structure enables the proof; the rate need not transfer, though the method does wherever a closed reporter series exists.

6 Data and code

The scan scripts, the findings file with per-finding context and adjudication fields, the denominator tables, and the series index are released with the corpus [6]. Every number in this paper regenerates from the released artifacts; none is transcribed. [pending: release manifest, DOI]

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